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New Classes of Perfect Maps I

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The existence and construction of *perfect maps*, also known as de Bruijn arrays or de Bruijn tori, is considered. A c-ary $(r, s; u, v)$ perfect map is a two-dimensional periodic array with periods r and s and symbols from an alphabet of size c with the property that every possible $u \times v$ array of symbols occurs exactly once in a period of the array. They generalise the well-known de Bruijn sequences. Simple necessary conditions on the parameters r, s, u, v for the existence of perfect maps are given. These conditions are shown to be sufficient when c is a power of a prime by constructing perfect maps for every allowed parameter set. This result will be applied in the second part to construct further c-ary perfect maps. © 1996 Academic Press, Inc.

1. INTRODUCTION

We begin with some formal definitions and go on to give a brief introduction to the subject of de Bruijn sequences and *perfect maps*.

Let $c \geqslant 2$ be a positive integer and let $\mathscr{A}$ be an alphabet of size c, that is, any set with c elements. We will be concerned with c-ary sequences and arrays, i.e., sequences and arrays whose terms come from such an alphabet $\mathscr{A}$. Throughout, p denotes a prime, and α a positive integer. We will often find it convenient to work with the alphabet $GF(p^\alpha)$, the finite field with p^α elements. We consider the elements of $GF(p)$ as being the integers modulo p, and will frequently represent the elements β of $GF(p^\alpha)$ as α-tuples $(b_0, \ldots, b_{\alpha-1})$ of elements of $GF(p)$.

A sequence $(s) = \ldots, s_{-1}, s_0, s_1, \ldots$ is said to *have period* t if it satisfies $s_i = s_{i+t}$ for every $i \in \mathbf{Z}$. Notice that t here need not be the least positive integer such that $s_i = s_{i+t}$ for every $i \in \mathbf{Z}$. If this is the case, then we refer to t as the *least period* of (s). We write $S_{\mathscr{A}}(t)$ for the set of sequences of

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period t with terms from $\mathscr{A}$. We may represent $(s) \in S_{\mathscr{A}}(t)$ by its *generating cycle*:

$$(s) = [s_0, s_1, ..., s_{t-1}].$$

Two sequences are said to be *equivalent* if one is just a shift of the other, or in terms of generating cycles, if the generating cycle of one is just a cyclic shift of the generating cycle of the other. Otherwise we refer to *inequivalent* sequences.

DEFINITION. A *c-ary span u de Bruijn sequence over alphabet* $\mathscr{A}$ is a sequence of period c^u with terms from $\mathscr{A}$ of size c with the property that for every u-tuple $(x_0, x_1, ..., x_{u-1}) \in \mathscr{A}^u$ there exists a unique i with $0 \leqslant i < c^u$ such that

$$(x_0, x_1, ..., x_{u-1}) = (s_i, s_{i+1}, ..., s_{i+u-1}).$$

Informally, sequence (s) is a c-ary span u de Bruijn sequence if every possible u-tuple of elements of $\mathscr{A}$ occurs once within a period of (s).

For each c and u, it is known that there exist $(c-1)!^{c^{u-1}} c^{c^{u-1}-u}$ inequivalent of c-ary span u de Bruijn sequences (see [5, 1]). The properties and applications of these sequences have been the subject of intensive study for some 40 years and there now exists an extensive and mature literature, the highlights being presented in [10]. More recent results and construction methods for c-ary de Bruijn sequences can be found in [7, 23, 27, 28]. Applications have been proposed for de Bruijn sequences and the related de Bruijn graphs in many fields of communications and coding, [2, 6, 11, 12, 26]. Interesting generalisations to sequences possessing alternative subsequence properties can be found in [3].

More recently, attention has been directed to higher dimensional versions of de Bruijn sequences, particularly in two dimensions. These are known variously as *perfect maps* [8, 21, 25], de Bruijn tori [4, 13] and de Bruijn arrays [9, 17]. We give a formal definition after introducing some more notation.

Let $A = (a_{i,j})$, $i, j \in \mathbf{Z}$ be an infinite two-dimensional array with period (r, s) over alphabet $\mathscr{A}$ of size c. That is,

$$a_{i,j} = a_{i+r,j} = a_{i,j+s} \qquad \text{for all} \quad i, j \in \mathbf{Z}, \qquad \text{where} \quad a_{i,j} \in \mathscr{A}.$$

It will be convenient to identify such an array A with the finite $r \times s$ array whose elements are $a_{i,j}$, $0 \leqslant i < r$, $0 \leqslant j < s$, which we call the *generating array for A*. The (u, v) subarray of A at position (x, y) is defined to be the $u \times v$ array whose entries are

$$a_{x+i,y+j}, \qquad 0 \leqslant i < u, \quad 0 \leqslant j < v.$$

DEFINITION. We say that A is a c-ary $(r, s; u, v)$ perfect map (PM) if every possible c-ary $u \times v$ array occurs as a (u, v) subarray of A in a unique position (x, y) with $0 \leqslant x < r$ and $0 \leqslant y < s$.

Perfect maps can also be used in two-dimensional position sensing and ranging applications akin to those outlined above for de Bruijn sequences—see, for example, [2, 20, 25]—and have applications in scrambling and various kinds of mask configurations (see [8]). In this paper and its successor we concentrate on combinatorial aspects of such arrays; in particular we study the existence question for perfect maps. In this direction, the following necessary conditions are easily established.

RESULT 1.1. *Suppose there exists a c-ary $(r, s; u, v)$ PM. Then*

(i) $rs = c^{uv}$,

(ii) $r > u$ *or* $r = u = 1$,

(iii) $s > v$ *or* $s = v = 1$.

The cases where $r = u = 1$ and $s = v = 1$ correspond to de Bruijn sequences and need not concern us further here.

As a simple consequence of Result 1.1(i), any prime p dividing r or s must also divide c. We suppose that c has prime factorisation

$$c = \prod_{i=1}^{n} p_i^{\alpha_i}$$

and write

$$r = \prod_{i=1}^{n} p_i^{k_i}, \qquad s = \prod_{i=1}^{n} p_i^{\alpha_i uv - k_i},$$

where $0 \leqslant k_i \leqslant \alpha_i uv$.

We conjecture that the conditions of Result 1.1 are in fact sufficient for the existence of perfect maps (see also [13, Conjecture 5.1], which is rather weaker). Indeed, almost all the work on perfect maps to date can be regarded as proving this conjecture in special cases. We summarise that work now. A series of papers [8, 9, 17, 20] culminating in [21] established the sufficiency of the conditions of Result 1.1 in the binary case, $c = 2$. Recently the enumeration of binary perfect maps has been considered [24]. A construction for c-ary maps generalising that of [17] can be found in [4]. The existence question for c-ary square perfect maps (i.e., PMs having $r = s$ and $u = v$) was partially settled in [9, 13, 14].

RESULT 1.2. *Suppose there exists a c-ary* $(r, r; u, u)$ *PM. Then either* u *is even or* c *is a perfect square. Conversely, if* u *is even or* c *is a perfect square, and one or more of the conditions*

(i) $c=2$ [9, *Theorem* 1.5],

(ii) $u=2$ [14],

(iii) c *is even and* $u \geqslant 10$ [13, *Theorem* 1.3.],

(iv) c *is odd* [13, *Theorem* 1.3.],

hold, then there exists a c-ary $(r, r; u, u)$ *PM.*

Another (much larger) class of parameter sets, analogous to those of [8], were considered in [22].

RESULT 1.3 [22, Theorem 6.2]. *Suppose* $c \geqslant 2$ *has prime factorisation*

$$c = \prod_{i=1}^{n} p_i^{\alpha_i}$$

as above and that t *has prime factorisation*

$$t = \prod_{i=1}^{n} p_i^{k_i}$$

with

$$p_i^{k_i} \leqslant p_i^{\alpha_i u}, \qquad 1 \leqslant i \leqslant n$$

and

$$u+1 \leqslant p_j^{k_j} \qquad \text{for some} \quad 1 \leqslant j \leqslant n.$$

Suppose further that v *is a positive integer such that when* $v=2$ *and* t *is even, we have* $2t \mid c^u$. *Then there exists a c-ary* $(t, c^{uv}/t; u, v)$ *perfect map.*

We shall have more to say about the method used in proving Result 1.3 in the sections to follow.

Taken together, the existence results outlined above are still far from providing a construction for perfect maps for all the parameter sets allowed by Result 1.1. Our aim in these papers is to move closer to the proof of our conjecture that the conditions of Result 1.1 are sufficient. In this first part, we prove the following theorem which forms a basis for the results in the second part.

THEOREM 1.4. *The conditions of Result* 1.1 *are sufficient for the existence of a c-ary* $(r, s; u, v)$ *PM when* c *is a power of a prime.*

Notice that when $c=p^{\alpha}$ is a power of a prime, we have $rs=p^{\alpha uv}$ so that $r=p^k$ and $s=p^{\alpha uv-k}$ for some k. We will frequently refer to p^{α}-ary $(p^k, p^{\alpha uv-k}; u, v)$ PMs in this case.

Theorem 1.4 is already true in the case $c=2$, according to the result of [21]. We will use techniques similar to those used in that paper, but generalised to $GF(p^\alpha)$. At one point we will also use the result of [14] on square PMs with $u=v=2$. Finally, we often take advantage of the obvious fact that the transpose of a c-ary $(r, s; u, v)$ PM is a c-ary $(s, r; v, u)$ PM.

2. Notation and Basic Results

In this section, we outline some basic results on the linear complexity of sequences over $GF(p^\alpha)$ having special periods and on properties of the generalised Lempel homomorphism. These results will be applied in the construction of PMs in the following sections. Proofs can be found in [22].

Definition. Suppose that for some $c_0, c_1, \ldots, c_{u-1} \in GF(p^\alpha)$ the sequence $(s) = \ldots, s_{-1}, s_0, s_1, \ldots$ of terms of $GF(p^\alpha)$ satisfies

$$s_{i+u} + c_0 s_i + c_1 s_{i+1} + \cdots + c_{u-1} s_{i+u-1} = 0 \qquad \text{for all} \quad i \in \mathbf{Z},$$

that is, a *linear recurrence relation of degree u.* Then the polynomial $g(X)$ given by

$$g(X) = c_0 + c_1 X + c_2 X^2 + \cdots + c_{u-1} X^{u-1} + X^u$$

is called the *characteristic polynomial* of the corresponding linear recurrence.

Any periodic sequence satisfies some linear recurrence, for if $(s) \in S_{GF(p^\alpha)}(t)$ then (s) satisfies $s_{i+t} = s_i$ for all $i \in \mathbf{Z}$, a linear recurrence of degree t. We have the following result and definitions.

Result 2.1 [16, Theorem 6.42]. *Let $(s) \in S_{GF(p^\alpha)}(t)$. Then there exists a uniquely determined monic polynomial $m(X) \in GF(p^\alpha)[X]$ having the following property: a polynomial $g(X) \in GF(p^\alpha)[X]$ is the characteristic polynomial of a linear recurrence generating (s) if and only if $m(X)$ divides $g(X)$.*

Definition. The polynomial $m(X)$ in Result 2.1 is called the *minimal polynomial* of (s).

Definition. Suppose $(s) \in S_{GF(p^\alpha)}(t)$. Then the *linear complexity* of (s), denoted $c(s)$, is the least positive integer c such that (s) satisfies a linear recurrence of degree c, that is, the degree of the minimal polynomial $m(X)$ of (s).

DEFINITION. Let $(s) = ..., s_{-1}, s_0, s_1, ...$ be an arbitrary sequence over $GF(p^\alpha)$. We define the *left shift operator* E acting on the sequence (s) by defining $E(s)$ to be the sequence whose ith term is s_{i+1}.

If $g(X)$ is a characteristic polynomial for a linear recurrence generating (s), we have, for all $i \in \mathbf{Z}$,

$$g(E)(s) = [0],$$

the all-zero sequence. When $(s) \in S_{GF(p^\alpha)}(t)$, $(E^t - 1)(s) = [0]$ and so $m(E) \mid E^t - 1$. Further when $t = p^k$, we have $E^t - 1 = (E-1)^t$, since $GF(p^\alpha)$ has characteristic p. Then the minimal polynomial of (s) is simply $m(E) = (E-1)^{c(s)}$ with the following consequence.

RESULT 2.2 [22, Theorem 2.5.]. *$(s) \in S_{GF(p^\alpha)}(p^k)$ has linear complexity c if and only if there exists a nonzero $a \in GF(p^\alpha)$ such that*

$$(E-1)^{c-1}(s) = [a].$$

DEFINITION. We define the *weight* of a sequence $(s) \in S_{GF(p^\alpha)}(t)$, denoted by $\mathrm{wt}(s)$, to be the sum $s_0 + s_1 + \cdots + s_{t-1}$. Note that the period t here need not be the least period of (s) and the sum is evaluated over the field $GF(p^\alpha)$.

RESULT 2.3 [22, Corollary 2.6]. *Suppose (s) has least period p^k and linear complexity c. Then*

$$p^{k-1} + 1 \leqslant c \leqslant p^k$$

and $c = p^k$ if and only if (s) has nonzero-weight.

By the discussion above on characteristic polynomials of sequences of period p^k, we see that a sequence in $S_{GF(p^\alpha)}(p^k)$ has a minimal polynomial of the form $m(E) = (E-1)^{c(s)}$ so that the coefficients of $m(E)$ are elements of $GF(p)$. Then, in calculating linear complexities of sequences of period p^k over $GF(p^\alpha)$, we may work component-wise. If $(s) = ..., s_{-1}, s_0, s_1, ...$ and $s_i = (s_i^1, ..., s_i^\alpha)$, where $s_i^j \in GF(p)$, then we have α *component sequences* over $GF(p)$,

$$(s)^j = ..., s_{-1}^j, s_0^j, s_1^j, ...,$$

each of period p^k, with

$$(s) = ((s)^1, ..., (s)^\alpha).$$

In the notation of Result 2.2,

$$(E-1)^{c-1}(s) = [a] \qquad \text{for some} \quad a \neq 0$$

if and only if

$$(E-1)^{c-1}(s)^j = [a_j] \qquad \text{for some} \quad a = (a_1, ..., a_\alpha) \neq (0, ..., 0).$$

Then it is clear that

$$c(s) = \max\{c((s)^j) : 1 \leqslant j \leqslant \alpha\}.$$

We now outline some relevant properties of the operator $(E-1)^{-1}$. These properties are related to the fact that $E-1$ is essentially the generalisation of Lempel's homomorphism [15] to an arbitrary finite field (see [22, 23] for further discussion).

We state two results, whose proofs are analogous to those of Theorems 3.3 and 3.4 of [22].

THEOREM 2.4. *Suppose* $(s) \in S_{GF(p^\alpha)}(t)$ *has zero-weight and generating cycle* $[s_0, ..., s_{t-1}]$. *Then the set of preimages of* (s) *under* $(E-1)^{-1}$, *denoted by* $(E-1)^{-1}(s)$, *consists of the* p^α *inequivalent sequences of period* t *with generating cycles*

$$(E-1)^{-1}(s) = \left\{ \left[a, a+s_0, a+s_0+s_1, ..., a+\sum_{i=0}^{t-2} s_i \right] : a \in GF(p^\alpha) \right\},$$

where the sum is evaluated over $GF(p^\alpha)$.

Notice here that, for every $a \in GF(p^\alpha)$, there is a sequence in the set $(E-1)^{-1}(s)$ whose generating cycle begins with a.

THEOREM 2.5. *Suppose* $(s) \in S_{GF(p^\alpha)}(t)$ *has nonzero-weight and generating cycle* $[s_0, ..., s_{t-1}]$. *Then the set of preimages of* (s) *under* $(E-1)^{-1}$, *denoted by* $(E-1)^{-1}(s)$, *consists of the* p^α *sequences of period* pt *with generating cycles given by*

$$\begin{aligned}(E-1)^{-1}(s) = \Bigg\{ \Bigg(\Bigg[& a, a+s_0, a+s_0+s_1, ..., a+\sum_{i=0}^{t-2} s_i, \\ & a+\mathrm{wt}(s), a+\mathrm{wt}(s)+s_0, ..., a+\mathrm{wt}(s)+\sum_{i=0}^{t-2} s_i, \\ & a+2\mathrm{wt}(s), a+2\mathrm{wt}(s)+s_0, ..., a+2\mathrm{wt}(s)+\sum_{i=0}^{t-2} s_i, \\ & \vdots \\ & a+(p-1)\,\mathrm{wt}(s), a+(p-1)\,\mathrm{wt}(s)+s_0, ..., \\ & a+(p-1)\,\mathrm{wt}(s)+\sum_{i=0}^{t-2} s_i \Bigg] : a \in GF(p^\alpha) \Bigg) \Bigg\},\end{aligned}$$

where the sums are evaluated over $GF(p^\alpha)$.

Note that in Theorem 2.5, the sequences whose generating cycles begin with the terms $a, a+\mathrm{wt}(s), ..., a+(p-1)\,\mathrm{wt}(s)$ are all equivalent to one another. Thus in this case, $(E-1)^{-1}(s)$ contains $p^{\alpha-1}$ inequivalent sequences.

By Result 2.2, when t is a power of p, say $t=p^k$, applying $E-1$ to a sequence (s) produces a sequence having linear complexity equal to $c(s)-1$ while the sequences in $(E-1)^{-1}(s)$ have linear complexity equal to $c(s)+1$. When the weight of (s) is zero, $c(s)<p^k$ and applying Theorem 2.4, $(E-1)^{-1}(s)$ consists of a set of p^α sequences of period p^k and linear complexity $c(s)+1\leqslant p^k$. When the weight of (s) is nonzero (and so the linear complexity of (s) is p^k), Theorem 2.5 implies that $(E-1)^{-1}(s)$ consists of $p^{\alpha-1}$ inequivalent sequences of period p^{k+1} and linear complexity $c(s)+1=p^k+1$.

We note the effect of $E-1$ and $(E-1)^{-1}$ on the subsequences of (s) in the next two lemmas, whose proofs follow easily from Theorem 2.4 and Theorem 2.5.

LEMMA 2.6. *Let* $(s)\in S_{GF(p^\alpha)}(t)$. *The subsequences of* $(E-1)(s)$ *of length* $u-1$ *are of the form* $(s_{i+1}-s_i, ..., s_{i+u-1}-s_{i+u-2})$, *where* $(s_i, s_{i+1}, ..., s_{i+u-1})$ *is a subsequence of* (s) *of length* u.

LEMMA 2.7. *Let* $(s)\in S_{GF(p^\alpha)}(t)$. *The subsequences of the sequences of* $(E-1)^{-1}(s)$ *of length* $u+1$ *are of the form* $(a, a+s_i, ..., a+s_i+s_{i+1}+\cdots+s_{i+u-1})$, *where* $a\in GF(p^\alpha)$ *is arbitrary and* $(s_i, s_{i+1}, ..., s_{i+u-1})$ *is a subsequence of* (s) *of length* u.

Thus when we apply the operator $(E-1)^{-1}$ to a sequence of known linear complexity, we can predict the period of the resulting sequences and have information on the form taken by their subsequences.

3 Zero-Weight Perfect Factors

DEFINITION. A (u, t, c) perfect factor (PF) over alphabet $\mathscr{A}$ of size c is a set of sequences of least period t with symbols from $\mathscr{A}$ having the property that each of the c^u different u-tuples of $\mathscr{A}^u$ arises exactly once as a subsequence in a period of a sequence of the set.

We have the following necessary conditions for the existence of perfect factors.

RESULT 3.1 [22, Theorem 1.2]. *Suppose there exists a* (u, t, c) *PF. Then* $t \mid c^u$ *and either* $u=t=1$ *or* $u+1\leqslant t\leqslant c^u$.

Perfect factors are of particular importance in the construction of perfect maps.

RESULT 3.2 [8, Construction A; 22, Theorem 6.1]. *Suppose there exists a* (u, t, c) *PF over* $\mathscr{A}$ *and* v *is a positive integer such that when* $v = 2$ *and* t *is even, then* $2t \mid c^u$. *Then there exists a* c*-ary* $(t, c^{uv}/t; u, v)$ *perfect map over* $\mathscr{A}$.

The columns of the perfect map of Result 3.2 are formed from shifts of the sequences of an appropriate perfect factor. In the case where $\mathscr{A} = GF(p^\alpha)$ and every sequence of the perfect factor has zero-weight, we can deduce that every column of the resulting PM (regarded as sequences of period t) also has zero-weight.

There is a simple method for choosing the shifts and ordering of factor sequences in Result 3.2 when $c = p^\alpha$ is a power of a prime and $v \geqslant 2$. This construction was alluded to in [22, Theorem 6.1] as a generalisation of [20, Construction 3.2]. Since we will make frequent use of it, we describe the method in detail below.

Let $c = p^\alpha$ be a power of a prime, and Γ a (u, p^k, p^α) PF with sequences

$$(s)^0, (s)^1, \ldots, (s)^{p^{\alpha u - k} - 1}.$$

Notice that when $p = 2$, the condition of Result 3.2, "when $v = 2$ and t is even, $2t \mid c^u$," becomes simply $k \leqslant \alpha u - 1$ when $v = 2$. Let (r) be a $p^{\alpha u - k}$-ary span v de Bruijn sequence over $\mathbf{Z}_{p^{\alpha u - k}}$. Let (d) be a p^k-ary span $v - 1$ de Bruijn sequence over $\mathbf{Z}_{p^k}$ whose generating cycle begins with $v - 1$ zeros and let $(d)'$ denote the sequence of period $p^{k(v-1)} - 1$ derived from (d) by deleting one of the initial zeros so $(d)' = [d_1, d_2, \ldots, d_{p^{k(v-1)}-1}]$. Then we define sequence (a) to have generating cycle

$$(a) = [0, 0, \ldots, 0, d_1, d_2, \ldots, d_{p^{k(v-1)}-1}, \ldots, d_1, d_2, \ldots, d_{p^{k(v-1)}-1}],$$

consisting of $p^{\alpha uv - kv}$ zeros, followed by $p^{\alpha uv - kv}$ copies of the generating cycle for $(d)'$. Finally, we define (w) by

$$w_0 = 0, \qquad w_i = \sum_{j=0}^{i-1} a_j \bmod p^k \qquad \text{for} \quad i \geqslant 1.$$

Here the sum is evaluated over $\mathbf{Z}_{p^k}$.

We specify that array A has column i equal to $E^{w_i}(s)^{r_i}$ and claim that A is indeed a p^α-ary $(p^k, p^{\alpha u - k}; u, v)$ PM. This can be proved in exactly the same way as Theorem 3.4 of [20], the crucial points being that the period of $(d)'$ and (r) are co-prime and that (w) has period $p^{\alpha uv - k}$.

In the case $v = 1$, a $(p^k, p^{\alpha u - k}; u, 1)$ PM can be obtained by arranging the sequences of the (u, p^k, p^α) PF as the columns of an array in any order, with any shifts.

As a partial converse to Result 3.2 we have the following

RESULT 3.3 [22, Theorem 4.3.]. *Suppose there exists a c-ary* $(r, s; u, v)$ *perfect map* A *over alphabet* $\mathscr{A}$. *Then there exists a* (u, r, c^v) *PF* Γ *over alphabet* $\mathscr{A}^v$.

Again we consider the case where $\mathscr{A} = GF(p^\alpha)$. According to the proof of Result 3.3, the component sequences of the sequences of the perfect factor Γ arise as columns of the perfect map A. Thus if each column of A has zero-weight, then so do the component sequences of each sequence of Γ, and hence the sequences of Γ themselves have this property.

Constructions for binary perfect factors were given in [8]. The case of nonbinary alphabets was considered in [22], where the conditions of Result 3.1 were shown to be sufficient in the case where c is a power of a prime and a class of PFs over nonprime-power alphabets was constructed. Further constructions for general c-ary alphabets can be found in [18, 19]. Perfect factors from [22] and [18] were used with Result 3.2 to prove Result 1.3. We note that the general existence question for PFs is not yet settled. Although the columns of a PM with $v = 1$ are just the sequences of a PF (see Result 3.3), it does seem necessary to solve this first existence problem before our conjecture on the sufficiency of the conditions of Result 1.1 can be proved.

For our purposes, we require perfect factors over $GF(p^\alpha)$ whose sequences all have zero-weight. We will speak of *zero-weight perfect factors*. Serendipitously, such factors are provided by the constructions of [22].

THEOREM 3.4. *Let* p *be a prime and* $\alpha \geqslant 1$. *For every* k *with* $u + 1 < p^k \leqslant p^{\alpha u}$, *there exists a zero-weight* (u, p^k, p^α) *PF over* $GF(p^\alpha)$.

Proof. A PF with $u = 1$ over $GF(p^\alpha)$ can be realised by using a partition of $GF(p^\alpha)$ into sets of equal size. When $p = 2$ and $\alpha = 1$, no k satisfies the conditions of the theorem. So for $p = 2$ we can assume $\alpha \geqslant 2$. We must have $k \geqslant 2$. Consider the partition of $GF(2^\alpha)$ into the $2^{\alpha-2}$ sets $A_x = \{(0, 0, x), (0, 1, x), (1, 0, x), (1, 0, x), (1, 1, x)\}$, where $x \in GF(2^{\alpha-2})$ if $\alpha \geqslant 3$ and x its omitted if $\alpha = 2$. The sum of the elements of each set A_x is zero. Taking appropriate unions of sets A_x, we obtain partitions of $GF(2^\alpha)$ into sets of size 2^k whose elements also sum to zero, for each $2 \leqslant k \leqslant \alpha$. These sets can be used to form the sequences of a $(u, 2^k, 2^\alpha)$ PF. When $p \geqslant 3$, similar considerations applied to the sets $A_x = \{(0, x), (1, x), \ldots, (p-1, x)\}$, where $x \in GF(p^{\alpha-1})$, will give PFs with the desired properties.

To prove the result for $u \geqslant 2$, we follow the constructions of PFs given in [22]. In particular we examine the linear complexities of the sequences of PFs constructed in [22, Theorems 3.6 and 4.6]. Theorem 3.6 of [22]

covers the possible parameter sets in the case $\alpha = 1$ and its proof was split into three further cases:

(i) When $u+1 < p^k \leqslant pu$, Corollary 2.8 of [22] was used to obtain an appropriate (u, p^k, p) PF whose sequences have linear complexity $u+1 < p^k$ and consequently zero-weight.

(ii) When $p^k = p^u$, the PF is just a de Bruijn sequence, and for $u \geqslant 2$ a span u de Bruijn sequence over $GF(p^\alpha)$ has zero-weight.

(iii) Otherwise $pu \leqslant p^k \leqslant p^{u-1}$. In the proof of Theorem 3.6 of [22], a (u, p^k, p^α) PF with sequences having linear complexity $p^{k-1} + u - k$ was produced for each such k. For $pu \leqslant p^k$, we have $u - k \leqslant p^{k-1} - k$ and so the sequences of this PF have linear complexity strictly less than p^k and, hence, zero-weight.

Theorem 4.6 of [22] deals with the possible parameter sets (u, p^k, p^α) when α is greater than one, and again several cases were considered in its proof:

(i) When $u+1 < p^k \leqslant p^u$, the result of [22, Corollary 4.2] was applied to a (u, p^k, p) PF Γ to obtain the desired (u, p^k, p^α) PF Γ' with sequences having linear complexities bounded by the maximum of the linear complexities of the sequences of Γ. From our discussion of the case $\alpha = 1$, the sequences of Γ can be chosen to have linear complexities that are less than p^k, so the same is true of the sequences of Γ'.

(ii) When $p^u < p^k < (u+1)p^u$, the proof of [22, Theorem 4.6] differed for $p = 2$ and $p \geqslant 3$. We consider first the case $p \geqslant 3$. In the final stage of the proof $s < u+1$ applications of $(E-1)^{-1}$ were made to a $(u-s, p^k, p^\alpha)$ PF whose sequences have linear complexity at most $p^k - (u-1)$ to obtain a set of sequences forming a (u, p^k, p^α) PF. Consequently, the sequences of the final PF have linear complexities at most $p^k - 1$ and hence they have zero-weight.

Secondly, for $p = 2$, the proof of [22, Theorem 4.6] made use of Result 3.3 applied to the binary perfect maps of [21]. It is easily checked that the argument used in the proof of [22, Theorem 4.6] for $p \geqslant 3$ is also valid for $p = 2$ and $u \geqslant 5$, and the above result on linear complexity holds mutatis mutandis. Otherwise, we have $2 \leqslant u \leqslant 4$, $\alpha \geqslant 2$, and $2^u < 2^k < (u+1)2^u$ and we require a $(u, 2^k, 2^\alpha)$ PF with zero-weight sequences. In the case $u = 2$, $k = 3$, $\alpha = 2$ such a PF is provided by [22, Example 1.1]. In the other cases, the required PF can be obtained by applying Result 3.3 to a binary $(2^k, 2^{\alpha u - k}; u, \alpha)$ PM having zero-weight columns. To show that such a PM exists requires a tedious application of the results of [21] and is relegated to Appendix A.

(iii) Finally, when $(u+1)p^u \leqslant p^k \leqslant p^{\alpha u}$, Lemma 4.4 of [22] was applied to a $(u, p^{k-u}, p^{\alpha-1})$ PF to obtain a (u, p^k, p^α) PF whose sequences have linear complexities at most $p^k - p^{k-\alpha u}$. Thus the sequences of the final PF have zero-weight. ▮

Notice that Theorem 3.4 does not provide a zero-weight PF when $u+1=p^k$. In fact a zero-weight PF with these parameters cannot exist, as is easily shown by considering the factor sequence containing the u-tuple $(0, 0, ..., 0)$.

As a consequence of Theorem 3.4 and Result 3.2 we have the following.

THEOREM 3.5. *Let p be a prime and $\alpha \geqslant 1$. For every $u \geqslant 1$, $v \geqslant 1$, and k with*

$$u+1 < p^k \leqslant p^{\alpha u - 1} \quad \text{when} \quad p=2;\ v=2,$$
$$u+1 < p^k \leqslant p^{\alpha u} \quad \text{otherwise},$$

there exists a p^α-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM over $GF(p^\alpha)$ whose columns all have zero-weight.

4. A Recursive Perfect Map Construction

We give a construction for PMs which generalises Construction 5.1 of [9]. It produces a p^α-ary $(p^k, p^{\alpha(u+1)v-k}; u+1, v)$ PM from a p^α-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM whose columns have zero-weight. In what follows, we regard both the columns and rows of our PMs as being periodic sequences. We also write $s = p^{\alpha uv-k}$.

Let $(d) = [d_0, d_1, ..., d_{p^{\alpha v}-1}]$ be a span v de Bruijn sequence over $GF(p^\alpha)$ whose generating cycle begins with the v-tuple $(0, 0, ..., 0)$ and let $(d)'$ denote the sequence $[d_1, d_2, ..., d_{p^{\alpha v}-1}]$ of period $p^{\alpha v}-1$. We form the sequence (e) of period $p^{\alpha v}s$ by defining its generating cycle to be the concatenation of s zeroes and s copies of the generating cycle for $(d)'$:

$$(e) = [0, 0, ..., 0, d_1, d_2, ..., d_{p^{\alpha v}-1}, ..., d_1, d_2, ..., d_{p^{\alpha v}-1}].$$

Let A be a p^α-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM over $GF(p^\alpha)$ with every column having zero-weight (alternatively, linear complexity less than p^k). We denote column j of A by $(a)^j$ and define an array B by taking column j of B to be the sequence of the set $(E-1)^{-1}(a)^j$ whose generating cycle begins with e_j.

THEOREM 4.1. *Let A and (e) be as above and suppose B is obtained using the construction just described. Then B is a $(p^k, p^{\alpha(u+1)v-k}; u+1, v)$ PM over $GF(p^\alpha)$ and*

(i) *if every column of A has linear complexity less than or equal to c, then every column of B has linear complexity less than or equal to $c+1$,*

(ii) *the rows of B all have equal linear complexity that is less than or equal to*

$$p^{\alpha(u+1)v-k}-p^{\alpha uv-k}.$$

Proof. The proof that B is a PM with the stated parameters is entirely analogous to the proof of [9, Construction 5.1]. For an alternative version, see [23, Chap. 5]. The claims on linear complexity are verified in a similar way to those of [21, Lemma 3.2, Lemma 3.3]. ∎

It is important to note that the method of Theorem 4.1 can also be applied in the case where A has zero-weight rows to obtain a $(p^{k+\alpha v}, p^{\alpha uv-k}; u, v+1)$ PM whose columns have linear complexities that are bounded by an expression similar to that of Theorem 4.1(ii). For example, in view of their linear complexities, the rows of the PM B in Theorem 4.1 have zero-weight. So the construction of Theorem 4.1 can be applied to the rows of B; one such application produces a PM whose rows have linear complexities increased by one over those of B. From the bound of Theorem 4.1(ii), we see that up to $p^{\alpha uv-k}$ applications of Theorem 4.1 can be made to the rows of B, with each successive application resulting in a PM with zero-weight columns. This process can then be begun afresh on the columns of the resulting maps if desired. In this way, we can recursively construct a large family of perfect maps.

We will show that the repeated use of Theorem 4.1 in this way, in combination with Theorem 3.5, is powerful enough to prove Theorem 1.4. Our method is to construct a basic class of perfect maps over each $GF(p^\alpha)$ having rows with suitably low linear complexities. The details of this construction vary slightly in the cases $p \geqslant 3$ and $p=2$ and so we treat them separately in Sections 5 and 6. Then in Section 7, Theorem 4.1 is repeatedly applied to the rows of PMs from this basic class to obtain p^α-ary PMs for every possible parameter set needed to prove Theorem 1.4.

We immediately apply Theorem 4.1 to the perfect maps of Theorem 3.5 to obtain the beginnings of our basic class.

THEOREM 4.2. *Let p be a prime and $\alpha \geqslant 1$. For every $u \geqslant 2$, $v \geqslant 1$ and k with*

$$u+1 \leqslant p^k \leqslant p^{\alpha(u-1)-1} \quad \text{when} \quad p=2;\ v=2,$$

$$u+1 \leqslant p^k \leqslant p^{\alpha(u-1)} \quad \text{otherwise},$$

there exists a p^α-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM over $GF(p^\alpha)$ whose rows have equal linear complexities that are less than or equal to $p^{\alpha uv-k}-p^{\alpha(u-1)v-k}$.

Proof. By Theorem 3.5 for every $u \geqslant 2$, $v \geqslant 1$ and k with

$$(u-1)+1 < p^k \leqslant p^{\alpha(u-1)-1} \quad \text{when} \quad p=2;\ v=2,$$
$$(u-1)+1 < p^k \leqslant p^{\alpha(u-1)} \quad \text{otherwise,}$$

there exists a $(p^k, p^{\alpha(u-1)v-k}; u-1, v)$ PM over $GF(p^\alpha)$ with zero-weight columns. Applying Theorem 4.1 to this PM gives a $(p^k, p^{\alpha uv-k}; u, v)$ PM whose rows have linear complexities that are bounded as in the statement of the theorem. ∎

5. The Basic Class for $p \geqslant 3$

In this section p will be an odd prime. We aim to prove the following theorem.

THEOREM 5.1. *Suppose $u \geqslant v \geqslant 2$ and k satisfies $u+1 \leqslant p^k < (u+1)p^{\alpha u}$, i.e., $\log_p(u+1) \leqslant k < \alpha u + \log_p(u+1)$. Then there exists a p^α-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM over $GF(p^\alpha)$ whose rows have linear complexities at most $p^{\alpha uv-k} - (u-v)$.*

Consider first k with $\log_p(u+1) \leqslant k \leqslant \alpha(u-1)$. For such k, by Theorem 4.2, there exists a $(p^k, p^{\alpha uv-k}; u, v)$ PM A with rows having linear complexities less than or equal to $p^{\alpha uv-k} - p^{\alpha(u-1)v-k}$. Now

$$\alpha(u-1)v-k \geqslant \alpha(u-1)v-\alpha(u-1) = \alpha(u-1)(v-1) \geqslant u-1$$

and for $u \geqslant v \geqslant 2$,

$$p^{u-1} \geqslant u-2 \geqslant u-v.$$

Thus the PM A has rows with linear complexities satisfying the requirement of Theorem 5.1 and for the remainder of this section we can assume that

$$\alpha(u-1) < k < \alpha u + \log_p(u+1). \tag{5.1}$$

We write $k = \alpha(u-1)+t$, where $1 \leqslant t < \alpha + \log_p(u+1)$, and define $s = \lceil k/2\alpha + 1/2 \rceil = \lceil u/2 + t/2\alpha \rceil$. We next establish some useful inequalities involving these parameters.

Now

$$\alpha sv - k = \alpha \left\lceil \frac{k}{2\alpha} + \frac{1}{2} \right\rceil v - k$$
$$\geqslant \alpha \left(\frac{k}{2\alpha} + \frac{1}{2} \right) v - k$$

$$= \frac{k}{2} v + \frac{1}{2} \alpha v - k$$

$$= k\left(\frac{1}{2} v - 1\right) + \frac{1}{2} \alpha v$$

But $v \geqslant 2$ and so

$$p^{\alpha s v - k} \geqslant p^{\alpha v/2} \geqslant p^{v/2} \geqslant v \qquad \text{for} \quad p \geqslant 3; v \geqslant 2. \tag{5.2}$$

For $u \geqslant 2$ and $1 \leqslant t < \alpha + \log_p(u+1)$ we have $s \geqslant 2$. Also, $s \geqslant k/2\alpha + 1/2$, so $2\alpha s \geqslant k + \alpha$, and, hence,

$$k - \alpha s \leqslant \alpha(s-1). \tag{5.3}$$

With these preliminaries, we now prove Theorem 5.1 for $u \geqslant 5$. This proof will illustrate some general principles that will be applied in the other cases.

Proof of Theorem 5.1 *for* $u \geqslant 5$. We claim that for k in the range of (5.1) and $u \geqslant 5$,

$$s + 1 \leqslant p^{k - \alpha s}. \tag{5.4}$$

To verify this claim, we consider separately the cases $u = 5$, $u = 6$, and $u \geqslant 7$. For $u \geqslant 7$, we can prove a general bound; we have

$$k - \alpha s = \alpha(u-1) - \alpha\left\lceil \frac{u}{2} + \frac{t}{2\alpha} \right\rceil + t$$

$$\geqslant \alpha(u-1) - \alpha\left(\frac{u}{2} + \frac{t}{2\alpha} + 1\right) + t$$

$$= \alpha\left(\frac{u}{2} - 2\right) + \frac{t}{2}$$

$$\geqslant \frac{u}{2} - \frac{3}{2} \qquad \text{since} \quad \alpha \geqslant 1; t \geqslant 1,$$

while

$$s + 1 \leqslant \frac{u}{2} + \frac{t}{2\alpha} + 2 \leqslant \frac{u}{2} + \frac{\log_3(u+1)}{2} + \frac{5}{2}.$$

It is easily checked that for $u \geqslant 7$,

$$\frac{u}{2} + \frac{\log_3(u+1)}{2} + \frac{5}{2} \leqslant 3^{u/2 - 3/2}$$

and so (5.4) holds for $u \geqslant 7$.

For $u = 5$, we have $1 \leqslant t < \alpha + \log_p(6)$ and so, since t is an integer, $1 \leqslant t \leqslant \alpha + 1$. Moreover, $s + 1 = \lceil 5/2 + t/2\alpha \rceil + 1$ while $p^{k-\alpha s} = p^{4\alpha - \alpha\lceil 5/2 + t/2\alpha \rceil + t}$. When $1 \leqslant t \leqslant \alpha$, $s + 1 = 4$, and $p^{k-\alpha s} = p^{\alpha + t} \geqslant p^2 \geqslant 9$, while then $t = \alpha + 1$, $s + 1 = 5$, and $p^{k-\alpha s} = p^{\alpha+1} \geqslant p^2 \geqslant 9$. In either case, $s + 1 \leqslant p^{k-\alpha s}$ and (5.4) holds for $u = 5$.

Similar reasoning applies in the case $u = 6$.

By Theorem 4.2, and inequalities (5.2), (5.3), and (5.4), there exists a $(p^{k-\alpha s}, p^{\alpha s v - k}; s, v-1)$ PM A over $GF(p^\alpha)$ with zero-weight rows. We apply Theorem 4.1 to the rows of A to deduce that there exists a $(p^k, p^{\alpha s v - k}; s, v)$ PM B over $GF(p^\alpha)$ whose columns have linear complexity less than or equal to $p^k - p^{k-\alpha s}$. Now $s = \lceil u/2 + t/2\alpha \rceil \geqslant u/2 + 1$ so $u - s \leqslant s - 1 < s + 1 \leqslant p^{k-\alpha s}$, where the last inequality follows from (5.4). Hence we can make $u - s$ applications of Theorem 4.1 to the columns of B to obtain a $(p^k, p^{\alpha u v - k}; u, v)$ PM C over $GF(p^\alpha)$ with zero-weight columns and whose rows have linear complexity less than or equal to $p^{\alpha u v - k} - p^{\alpha(u-1)v - k}$. To complete the proof we need only show that the rows of C have linear complexities satisfying the bound in the statement of Theorem 5.1. We have

$$\begin{aligned}
\alpha(u-1)\,v - k &\geqslant \alpha(u-1)\,v - \alpha u - \log_p(u+1) \qquad \text{(by (5.1))} \\
&= \alpha(u-1)(v-1) - \alpha - \log_p(u+1) \\
&\geqslant \alpha(u-2) - \log_p(u+1) \qquad \text{(since } v \geqslant 2\text{)} \\
&\geqslant u - 2 - \log_p(u+1) \qquad \text{(since } \alpha \geqslant 1\text{)},
\end{aligned}$$

while for $u \geqslant 5$ and $p \geqslant 3$, $p^{u-2-\log_p(u+1)} \geqslant u - 2 \geqslant u - v$. ∎

In order to prove Theorem 5.1 for $u = 2, 3, 4$ we first need to consider two exceptional cases. We will then apply a technique similar to that employed in the above proof.

LEMMA 5.2. *For every $\alpha \geqslant 1$, there exists a $(3^{3\alpha+1}, 3^{3\alpha-1}; 3, 2)$ PM over $GF(3^\alpha)$ having zero-weight rows and zero-weight columns.*

Proof. In the case $\alpha = 1$, a $(3^4, 3^2; 3, 2)$ PM over $GF(3)$ with zero-weight rows and zero-weight columns is required. The transpose of such a PM is constructed in Appendix B.

Suppose now that $\alpha \geqslant 2$. By Theorem 3.5 there exists a $(3^{\alpha-1}, 3^{\alpha+1}; 1, 2)$ PM over $GF(3^\alpha)$ having zero-weight columns. Applying Theorem 4.1 to the columns and transposing the resulting array, we get a $(3^{3\alpha+1}, 3^{\alpha-1}; 2, 3)$ PM over $GF(3^\alpha)$ whose columns have linear complexity less than or equal to $3^{3\alpha+1} - 3^{\alpha+1}$. One more application of Theorem 4.1 to the columns of this last PM produces a $(3^{3\alpha+1}, 3^{3\alpha-1}; 3, 2)$ PM over $GF(3^\alpha)$ whose columns have linear complexity less than or equal to $3^{3\alpha+1} - 3^{\alpha+1} + 1$ and whose rows have zero-weight. Since $\alpha \geqslant 2$, $3^{\alpha+1} - 1 \geqslant 8$ and the final PM also has zero-weight columns. ▮

LEMMA 5.3. *For every $\alpha \geqslant 1$, there exists a $(3^{3\alpha+1}, 3^{6\alpha-1}; 3, 3)$ PM over $GF(3^\alpha)$ having zero-weight columns.*

Proof. For $\alpha = 1$, we begin with a $(3^2, 3^2; 2, 2)$ PM over $GF(3)$ from Theorem 3.5. Its transpose is a PM with the same parameters having zero-weight rows. Applying Theorem 4.1 to the rows gives a $(3^4, 3^2; 2, 3)$ PM over $GF(3)$ whose columns have linear complexity less than or equal to $3^4 - 3^2$. Applying Theorem 4.1 to the columns of this PM gives a $(3^4, 3^5; 3, 3)$ PM over $GF(3)$ having zero-weight rows and columns.

For $\alpha \geqslant 2$, we commence with the transpose of a $(3^{\alpha-1}, 3; 1, 1)$ PM over $GF(3^\alpha)$ with zero-weight columns obtained from Theorem 3.5. The reader can check that successively applying Theorem 4.1 to the rows, columns, rows, and finally columns, we obtain a PM with the desired parameters having zero-weight columns. ▮

Proof of Theorem 5.1 *for* $2 \leqslant u \leqslant 4$. We begin with $u = 2$. From the statement of the theorem, the only v of interest is $v = 2$. Since $u = v$, we require to construct a $(p^k, p^{4\alpha-k}; 2, 2)$ PM over $GF(p^\alpha)$ for each k with $1 \leqslant k < 2\alpha + 1$, without a restriction on the linear complexities of rows. We can simply use Theorem 3.5.

Consider now $u = 3$. With notation as before, we have $k = \alpha(u-1) + t$, where $1 \leqslant t < \alpha + \log_p(u+1)$. When $p = 3$, $1 \leqslant t \leqslant \alpha + 1$, and when $p \geqslant 5$, $1 \leqslant t \leqslant \alpha$. We claim that, except in the case $t = \alpha + 1$ and $p = 3$, there exists a $(p^{k-\alpha s}, p^{\alpha sv-k}; s, v-1)$ PM A over $GF(p^\alpha)$ with zero-weight rows. From (5.2), (5.3), and Theorem 4.2, we need only show that $s + 1 \leqslant p^{k-\alpha s}$. With the exception above noted, we have $1 \leqslant t \leqslant \alpha$, $s + 1 = 3$, and $p^{k-\alpha s} = p^t \geqslant 3$ and our claim is true. We apply Theorem 4.1 to the rows of A to obtain a $(p^k, p^{\alpha sv-k}; s, v)$ PM B over $GF(p^\alpha)$ whose columns have linear complexity less than or equal to $p^k - p^{k-\alpha s}$. As before, $u - s \leqslant p^{k-\alpha s}$, and we can make $u - s$ applications of Theorem 4.1 to the columns of B to obtain a $(p^k, p^{3\alpha v-k}; 3, v)$ PM C over $GF(p^\alpha)$ whose rows have linear complexity less than or equal to $p^{3\alpha v-k} - p^{2\alpha v-k}$. Since $k \leqslant 3\alpha$ and $v \geqslant 2$, we have $p^{2\alpha v-k} \geqslant p^\alpha \geqslant 3 > u - v$. Thus C has rows satisfying the required bound on linear complexity.

In the exceptional case where $t = \alpha + 1$ and $p = 3$, the results of Lemma 5.2 and Lemma 5.3 provide PMs with the required parameters and properties.

Finally we consider $u = 4$. When $p = 3$, $1 \leqslant t \leqslant \alpha + 1$, and when $p \geqslant 5$, $1 \leqslant t \leqslant \alpha$. In both cases, $s + 1 = 4$. Also, as before, Eqs. (5.2) and (5.3) hold and $p^{k-\alpha s} = p^t$. Except in the case $p = 3$, $t = 1$, we have $s + 1 \leqslant p^{k-\alpha s}$, so that by Theorem 4.2 there exists a $(p^{k-\alpha s}, p^{\alpha sv-k}; s, v-1)$ PM A over $GF(p^\alpha)$ with zero-weight rows. The same procedure as in the case $u = 3$ can be used to generate a $(p^k, p^{4\alpha v-k}; 4, v)$ PM C over $GF(p^\alpha)$ whose rows have linear complexity less than or equal to $p^{4\alpha v-k} - p^{3\alpha v-k}$. Since $k \leqslant 4\alpha + 1$ and $v \geqslant 2$, we have $3\alpha v - k \geqslant 2\alpha - 1 \geqslant 1$ and so $p^{3\alpha v-k} \geqslant 3 > u - v$. Hence, the rows of C satisfy the required bound on linear complexity.

In the exceptional case of $p = 3$ and $t = 1$, we desire to construct $(3^{3\alpha+1}, 3^{5\alpha-1}; 4, 2)$, $(3^{3\alpha+1}, 3^{9\alpha-1}; 4, 3)$ and $(3^{3\alpha+1}, 3^{13\alpha-1}; 4, 4)$ PMs over $GF(3^\alpha)$ with the appropriate bounds on the linear complexities of rows. In the first two cases, PMs with the required properties can be obtained by applying Theorem 4.1 to the columns of the PMs of Lemma 5.2 and Lemma 5.3, respectively. PMs in the third case can be obtained from Theorem 3.5 (since we have no conditions on the linear complexity of rows when $u = v$). ∎

6. The Basic Class for $p = 2$

In this section we consider the construction of a basic class of PMs over $GF(p^\alpha)$ when $p = 2$ and $\alpha \geqslant 2$. We prove the following theorem, analogous to Theorem 5.1.

THEOREM 6.1. *Suppose $u \geqslant v \geqslant 2$ and k satisfies $u + 1 \leqslant 2^k < (u+1)\,2^{\alpha u}$, i.e., $\log_2(u+1) \leqslant k < \alpha u + \log_2(u+1)$. Then there exists a 2^α-ary $(2^k, 2^{\alpha uv-k}; u, v)$ PM over $GF(2^\alpha)$ whose rows have linear complexities at most $2^{\alpha uv-k} - (u - v)$.*

The technique we use to prove Theorem 6.1 is almost identical to that used in the previous section. However, we have to work somewhat harder because of the restriction on the class of PMs obtained in Theorem 3.5 in the case $p = 2$ and $v = 2$.

First, for k with

$$\log_2(u+1) \leqslant k < \alpha(u-1) \qquad \text{when} \quad v = 2,$$

$$\log_2(u+1) \leqslant k \leqslant \alpha(u-1) \qquad \text{when} \quad v \geqslant 3,$$

we have from Theorem 4.2 that there exists a $(2^k, 2^{\alpha uv-k}; u, v)$ PM A with rows having linear complexities less than or equal to $2^{\alpha uv-k}-2^{\alpha(u-1)v-k}$. Now

$$\alpha(u-1)v-k \geqslant \alpha(u-1)v-\alpha(u-1)=\alpha(u-1)(v-1)\geqslant u-1$$

and for $u \geqslant v \geqslant 2$,

$$2^{u-1} \geqslant u-2 \geqslant u-v.$$

Therefore A has rows with linear complexities satisfying the requirement of Theorem 6.1 and for the remainder of this section we need only consider k with

$$\begin{aligned} &\alpha(u-1) \leqslant k < \alpha u + \log_2(u+1) \quad \text{when} \quad v=2, \\ &\alpha(u-1) < k < \alpha u + \log_2(u+1) \quad \text{when} \quad v \geqslant 3. \end{aligned} \tag{6.1}$$

We write $k=\alpha(u-1)+t$, where $0 \leqslant t < \alpha + \log_2(u+1)$, and define

$$s = \begin{cases} \left\lceil \dfrac{u}{2} + \dfrac{t+1}{2\alpha} \right\rceil & \text{when} \quad v=2, \\ \left\lceil \dfrac{u}{2} + \dfrac{t}{2\alpha} \right\rceil & \text{when} \quad v \geqslant 3. \end{cases}$$

Clearly $s \geqslant 2$ for $u \geqslant 2$. An easy argument like that in the previous section shows that with this choice of s,

$$2^{\alpha sv-k} \geqslant v. \tag{6.2}$$

Similarly, we have

$$k-\alpha s \leqslant \begin{cases} \alpha(s-1)-1 & \text{when} \quad v=2, \\ \alpha(s-1) & \text{when} \quad v \geqslant 3. \end{cases}$$

We derive one final inequality that will be useful in this section, holding for k in the range of (6.1). We have

$$\begin{aligned} \alpha(u-1)v-k &\geqslant \alpha(u-1)v-\alpha u-\log_2(u+1) \\ &= \alpha(u-1)(v-1)-\alpha-\log_2(u+1) \\ &\geqslant \alpha(u-2)-\log_2(u+1) \qquad (\text{since } v \geqslant 2) \\ &\geqslant 2(u-2)-\log_2(u+1) \qquad (\text{since } \alpha \geqslant 2) \end{aligned}$$

and so for $u \geqslant 3$ and $u \geqslant v \geqslant 2$ it can be verified that

$$2^{\alpha(u-1)v-k} \geqslant 2^{2(u-2)-\log_2(u+1)} \geqslant u-2 \geqslant u-v. \tag{6.4}$$

Proof of Theorem 6.1 *for* $u \geqslant 7$. We claim that for k as in (6.1) and $u \geqslant 7$,

$$s+1 \leqslant 2^{k-\alpha s}. \tag{6.5}$$

We have

$$\begin{aligned} k-\alpha s &\geqslant u(\alpha-1)+t-\alpha\left\lceil \frac{u}{2}+\frac{t+1}{2\alpha}\right\rceil \\ &\geqslant \alpha(u-1)+t-\alpha\left(\frac{u}{2}+\frac{t+1}{2\alpha}+1\right) \\ &= \alpha\left(\frac{u}{2}-2\right)+\frac{t-1}{2} \\ &\geqslant u-\frac{9}{2} \qquad \text{since} \quad \alpha \geqslant 2,\ t \geqslant 0, \end{aligned}$$

while

$$s+1 \leqslant \left\lceil \frac{u}{2}+\frac{t+1}{2\alpha}\right\rceil + 1 \leqslant \frac{u}{2}+\frac{1}{4}(11+\log_2(u+1)).$$

It is easily verified from these bounds that (6.5) holds for $u \geqslant 8$. When $u=7$ and $v=2$, we have $0 \leqslant t < \alpha+3$ and

$$s+1 = \begin{cases} 4 & \text{for} \quad 0 \leqslant t \leqslant \alpha-1, \\ 5 & \text{for} \quad t=\alpha, \alpha+1, \alpha+2, \end{cases}$$

while

$$\begin{aligned} 2^{k-\alpha s} &= \begin{cases} 2^{2\alpha+t} & \text{for} \quad 0 \leqslant t \leqslant \alpha-1, \\ 2^{\alpha+t} & \text{for} \quad t=\alpha, \alpha+1, \alpha+2, \end{cases} \\ &\geqslant 2^{2\alpha} \\ &\geqslant 16 \end{aligned}$$

so that (6.5) holds for $u=7$ and $v=2$. That (6.5) holds for $u=7$ and $3 \leqslant v \leqslant 7$ can be verified in a similar fashion.

By Theorem 4.2 and inequalities (6.2), (6.3), and (6.5), there exists a $(2^{k-\alpha s}, 2^{\alpha s v-k}; s, v-1)$ PM A over $GF(2^\alpha)$ with zero-weight rows. We apply the result of Theorem 4.1 to the rows of A to deduce that there exists a $(2^k, 2^{\alpha s v-k}; s, v)$ PM B over $GF(p^\alpha)$ whose columns have linear complexity less than or equal to $2^k-2^{k-\alpha s}$. Now $s > \lceil u/2+t/2\alpha \rceil \geqslant u/2$ so $u-s \leqslant s < s+1 \leqslant 2^{k-\alpha s}$, where the last inequality follows from (6.5). Hence

we can make $u-s$ applications of Theorem 4.1 to the columns of B to obtain a $(2^k, 2^{\alpha uv-k}; u, v)$ PM C over $GF(2^\alpha)$ whose rows have linear complexity less than or equal to $2^{\alpha uv-k}-2^{\alpha(u-1)v-k}$. Moreover, from (6.4), we have $2^{\alpha(u-1)v-k} \geqslant u-v$ and so C has the required parameters and bound on linear complexities of its rows. ∎

In the next nine results, we construct a number of special PMs needed to prove Theorem 6.1 for u with $2 \leqslant u \leqslant 6$.

RESULT 6.2 [14]. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{2\alpha}, 2^{2\alpha}; 2, 2)$ PM.*

We can assume that the PMs of Result 6.2 are over the alphabet $GF(2^\alpha)$.

LEMMA 6.3. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{2\alpha}, 2^{4\alpha}; 3, 2)$ PM over $GF(2^\alpha)$ with zero-weight rows.*

Proof. Let (s) be a span 2 de Bruijn sequence over $GF(2^\alpha)$; (s) has zero-weight so, by Theorem 2.4, $(E-1)^{-1}(s)$ consists of the 2^α inequivalent sequences of period $2^{2\alpha}$ of the form

$$(s)^a = [a+t_0, a+t_1, \ldots, a+t_{2^{2\alpha}-1}], \qquad a \in GF(2\alpha)$$

for some $0 = t_0, t_1, \ldots, t_{2^{2\alpha}-1} \in GF(2^\alpha)$. Notice that we have indexed these sequences by the elements of $GF(2^\alpha)$. By [22, Lemma 4.5] the sequences constitute a $(3, 2^{2\alpha}, 2^\alpha)$ PF Γ. We use Γ in Result 3.2 to form our $(2^{2\alpha}, 2^{4\alpha}; 3, 2)$ PM. In the notation of the discussion following Result 3.2, we take (r) to be a span 2 de Bruijn sequence over $GF(2^\alpha)$ and define

$$(a) = [0, 0, \ldots, 0, 1, 2, \ldots, 2^{2\alpha}-1, \ldots, 1, 2, \ldots, 2^{2\alpha}-1],$$

consisting of $2^{2\alpha}$ zeros followed by $2^{2\alpha}$ copies of a $2^{2\alpha}$-ary span 1 de Bruijn sequence with leading term 0 removed. Then we define (w) by

$$w_0 = 0, \qquad w_i = \sum_{j=0}^{i-1} a_j \bmod 2^{2\alpha} \qquad \text{for} \quad i \geqslant 1$$

(where the sum is evaluated over $\mathbf{Z}_{2^{2\alpha}}$). Finally, we define array A to have column i equal to $E^{w_i}(s)^{r_i}$. Then A is a $(2^{2\alpha}, 2^{4\alpha}; 3, 2)$ PM.

We claim that the rows of A have zero-weight. Since $w_i = 0$ for $0 \leqslant i < 2^{2\alpha}$ and (r) is a span-2 de Bruijn sequence, the first $2^{2\alpha}$ columns of A consist of an ordering of the sequences of Γ with zero shifts, each sequence appearing 2^α times. Clearly the sum across these columns of A is zero. Now consider columns

$$2^{2\alpha} + i(2^{2\alpha}-1), \qquad 0 \leqslant i < 2^{2\alpha},$$

of A. Note that over $\mathbf{Z}_{2^{2\alpha}}$, we have

$$\sum_{j=1}^{2^{2\alpha}-1} j \bmod 2^{2\alpha} = 2^{2\alpha-1},$$

so that by the construction of (w) these columns of A are shifts of sequences of Γ by zero and $2^{2\alpha-1}$. Since (r) has period $2^{2\alpha}$, these columns are just $(s)^{r_0}, (s)^{r_2}, \ldots, (s)^{r_{2^{2\alpha}-2}}$ and $E^{2^{2\alpha-1}}(s)^{r_1}, E^{2^{2\alpha-1}}(s)^{r_3}, \ldots, E^{2^{2\alpha-1}}(s)^{r_{2^{2\alpha}-1}}$ in some order. From the form of the sequences $(s)^a$, it is clear that the sum across these columns of A is zero. A similar argument shows that, for each $1 \leqslant j < 2^{2\alpha}-1$, the sets of columns

$$2^{2\alpha} + j + i(2^{2\alpha}-1), \qquad 0 \leqslant i < 2^{2\alpha},$$

of A also sum to zero. Hence, the sum across all the columns of A is zero; i.e., every row of A has zero-weight. The result follows. ∎

LEMMA 6.4. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{2\alpha+1}, 2^{4\alpha-1}; 3, 2)$ PM over $GF(2^\alpha)$ with zero-weight rows.*

Proof. We begin by constructing a special class of $(2, 2^{2\alpha-1}, 2^\alpha)$ PFs over $GF(2^\alpha)$. Such a PF consists of two sequences of period $2^{2\alpha-1}$ which we will denote by (s) and (t). The special property that we demand of the PFs is that

$$s_k + s_{k+2^{2\alpha-2}} + t_k + t_{k+2^{2\alpha-2}} = 0 \qquad \text{for every } k.$$

Our proof that such a class exists is by induction. When $\alpha = 2$, it can be checked by hand that the PF of [22, Example 1.1] has the required property. Suppose that $\alpha \geqslant 3$ and suitable sequences (s) and (t) exist over $GF(2^{\alpha-1})$. These sequences will have period $2^{2\alpha-3}$. Let (b) denote the binary sequence of period $2^{2\alpha-1}$:

$$[0, 0, \ldots, 0, 0, 1, 1, 0, 1, 1, \ldots, 0, 1, 1]$$

consisting of $2^{2\alpha-3}$ zeros followed by $2^{2\alpha-3}$ copies of the sequence [011]. Let $(\bar{b})$ denote the term by term complement of (b). We define new sequences $(s)'$ and $(t)'$ over $GF(2^\alpha)$ by setting

$$s'_i = (s_i, b_i), \qquad t'_i = (t_i, \bar{b}_i)$$

so that the first $\alpha - 1$ components of s'_i and t'_i come from s_i and t_i, respectively. Clearly $(s)'$ and $(t)'$ have period $2^{2\alpha-1}$. Using similar

arguments as in the proof of [22, Lemma 4.5], it can be shown that $(s)'$, $(t)'$ constitute a $(2, 2^{2\alpha-1}, 2^{\alpha})$ PF. Moreover, for every k,

$$\begin{aligned} s'_k + s'_{k+2^{2\alpha-2}} + t'_k + t'_{k+2^{2\alpha-2}} &= (s_k, b_k) + (s_{k+2^{2\alpha-2}}, b_{k+2^{2\alpha-2}}) \\ &\quad + (t_k, \bar{b}_k) + (t_{k+2^{2\alpha-2}}, \bar{b}_{k+2^{2\alpha-2}}) \\ &= (s_k, b_k) + (s_k, b_{k+2^{2\alpha-2}}) + (t_k, \bar{b}_k) \\ &\quad + (t_k, \bar{b}_{k+2^{2\alpha-2}}) \\ &= ((0, \ldots, 0), 0) \end{aligned}$$

so that $(s)', (t)'$ have the required property. Thus the induction step is completed.

We will now use these PFs to construct a $(2^{2\alpha-1}, 2^{2\alpha+1}; 2, 2)$ PM over $GF(2^{\alpha})$ for each $\alpha \geqslant 2$. In the notation of the discussion following Result 3.2, we define $(r) = [0011]$ and

$$(a) = [0, 0, 0, 0, 1, 2, \ldots, 2^{2\alpha-1}-1, \ldots, 1, 2, \ldots, 2^{2\alpha-1}-1],$$

consisting of four zeros followed by four copies of a $2^{2\alpha-1}$-ary span 1 de Bruijn sequence with leading term 0 removed. Finally we define (w) by

$$w_0 = 0, \qquad w_i = \sum_{j=0}^{i-1} a_j \bmod 2^{2\alpha-1} \qquad \text{for} \quad i \geqslant 1,$$

where the sum is computed over $\mathbf{Z}_{2^{2\alpha-1}}$.

Let (s) and (t) be the sequences of the $(2, 2^{2\alpha-1}, 2^{\alpha})$ PF constructed above. We define array A to have column i equal to $E^{w_i}(s)$ if $r_i = 0$, and $E^{w_i}(t)$ if $r_i = 1$. A is a $(2^{2\alpha-1}, 2^{2\alpha+1}; 2, 2)$ PM and we claim that the rows of A have zero-weight. Note that columns 0, 1, 2, and 3 of A clearly sum to zero, since they consist of two copies each of (s) and (t) with zero shifts. For each $0 \leqslant j < 2^{2\alpha-1}-1$, consider the four columns

$$4+j, \quad 4+j+2^{2\alpha-1}-1, \quad 4+j+2(2^{2\alpha-1}-1), \quad 4+j+3(2^{2\alpha-1}-1)$$

of A. From the choice of (r), the construction of (w), and the fact that

$$\sum_{j=1}^{2^{2\alpha-1}-1} j \bmod 2^{2\alpha-1} = 2^{2\alpha-2},$$

these are either

$$E^i(s), \quad E^{i+2^{2\alpha-2}}(s), \quad E^i(t), \quad E^{i+2^{2\alpha-2}}(t)$$

in some order, for some i, or two copies each of $E^i(s)$ and $E^{i+2^{2\alpha-2}}(t)$ for some i. In either case, the four columns of A have sum zero—in the former

case by the special property of (s) and (t). Then the sum across all the columns of A is also zero and A has zero-weight rows.

We apply Theorem 4.1 to the rows of A and then transpose to obtain a $(2^{2\alpha+1}, 2^{4\alpha-1}; 3, 2)$ PM over $GF(2^\alpha)$ whose rows have zero-weight. ∎

LEMMA 6.5. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{3\alpha}, 2^{3\alpha}; 3, 2)$ PM over $GF(2^\alpha)$ with zero-weight rows and zero-weight columns.*

Proof. Let (s) be a span 1 de Bruijn sequence over $GF(2^\alpha)$, having zero-weight. By Theorem 2.4, the set $(E-1)^{-1}(s)$ consists of 2^α inequivalent sequences of period 2^α of the form

$$[a+t_0, a+t_1, \ldots, a+t_{2^\alpha-1}], \qquad a \in GF(2\alpha),$$

for some $0 = t_0, t_1, \ldots, t_{2^\alpha-1} \in GF(2^\alpha)$. By [22, Lemma 4.5] these sequences constitute a $(2, 2^\alpha, 2^\alpha)$ PF and by using these sequences as the columns we obtain a $(2^\alpha, 2^\alpha, 2, 1)$ PM A over $GF(2^\alpha)$. This PM has zero-weight rows, for the ith row has entries $a+t_i$, $a \in GF(2^\alpha)$. We now apply Theorem 4.1 once to the rows of A and then once to the columns of the resulting PM to obtain a $(2^{3\alpha}, 2^{3\alpha}; 3, 2)$ PM whose rows and columns all have zero-weight. ∎

LEMMA 6.6. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{3\alpha+1}, 2^{3\alpha-1}; 3, 2)$ PM over $GF(2^\alpha)$ with zero-weight rows and zero-weight columns.*

Proof. For the case $\alpha = 2$, see Appendix C. Otherwise, when $\alpha \geqslant 3$, by Theorem 3.5, there exists a $(2^{\alpha-1}, 2^{\alpha+1}; 1, 2)$ PM over $GF(2^\alpha)$ whose columns have zero-weight. We apply Theorem 4.1 to the columns to get a $(2^{\alpha-1}, 2^{3\alpha+1}; 2, 2)$ PM over $GF(2^\alpha)$ with rows having linear complexity at most $2^{3\alpha+1} - 2^{\alpha+1}$. Then we apply Theorem 4.1 to the rows to obtain a $(2^{3\alpha-1}, 2^{3\alpha+1}; 2, 3)$ PM with rows having linear complexity at most $2^{3\alpha+1} - 2^{\alpha+1} + 1$ and having zero-weight columns. Since $2^{\alpha+1} - 1 \geqslant 15$, this final PM has zero-weight rows. Its transpose therefore has the desired parameters and properties. ∎

LEMMA 6.7. *For every $\alpha \geqslant 2$, there exists a 2^α-ary $(2^{3\alpha+1}, 2^{6\alpha-1}; 3, 3)$ PM over $GF(2^\alpha)$ whose columns have zero-weight.*

Proof. For the case $\alpha = 2$, we begin with the $(2^3, 2^5; 2, 2)$ PM over $GF(2^2)$ with zero-weight rows constructed in Lemma 6.4. We apply Theorem 4.1 once to the rows of this PM and then once to the columns of the resulting PM to obtain a $(2^7, 2^{11}; 3, 3)$ PM over $GF(2^2)$ whose columns have linear complexity at most $2^7 - 2^3 + 1$ and, hence, zero-weight.

Otherwise, we have $\alpha \geqslant 3$. By transposing a PM from Theorem 3.5, we can get a $(2, 2^{\alpha-1}; 1, 1)$ PM over $GF(2^\alpha)$ whose rows have zero-weight. We apply Theorem 4.1 to the rows of this PM to get a $(2^{\alpha+1}, 2^{\alpha-1}; 1, 2)$ PM with zero-weight columns, then to the columns to get a $(2^{\alpha+1}, 2^{3\alpha-1}; 2, 2)$ PM with zero-weight rows and then to the rows again to get a $(2^{3\alpha+1}, 2^{3\alpha-1}; 2, 3)$ PM with columns having linear complexity at most $2^{3\alpha+1} - 2^{\alpha+1}$ and zero-weight rows. Finally, we apply Theorem 4.1 to the columns of this last PM to get a $(2^{3\alpha+1}, 2^{6\alpha-1}; 3, 3)$ PM with columns having linear complexity at most $2^{3\alpha+1} - 2^{\alpha+1} + 1$. Since $2^{\alpha+1} - 1 \geqslant 15$, our final PM has zero-weight columns. It also has the desired parameters. ▮

LEMMA 6.8. *For every* $\alpha \geqslant 2$, *there exist* 2^α*-ary* $(2^{3\alpha}, 2^{5\alpha}; 4, 2)$ *and* $(2^{3\alpha+1}, 2^{5\alpha-1}; 4, 2)$ *PMs over* $GF(2^\alpha)$, *having rows with linear complexities less than or equal to* $2^{5\alpha} - 2$ *and* $2^{5\alpha-1} - 2$, *respectively.*

Proof. The lemma follows directly on applying Theorem 4.1 to the columns of the PMs of Lemma 6.5 and Lemma 6.6. ▮

LEMMA 6.9. *For every* $\alpha \geqslant 2$, *there exists a* 2^α*-ary* $(2^{3\alpha+1}, 2^{9\alpha-1}; 4, 3)$ *PM over* $GF(2^\alpha)$ *with zero-weight rows.*

Proof. The lemma follows immediately on applying Theorem 4.1 to the PMs of Lemma 6.7. ▮

LEMMA 6.10. *There exist* $(2^{10}, 2^6; 4, 2)$, $(2^{10}, 2^{10}; 5, 2)$, *and* $(2^{10}, 2^{14}; 6, 2)$ *PMs over* $GF(2^2)$ *whose rows have linear complexities less than or equal to* $2^{10} - 2$, $2^{10} - 3$, *and* $2^{10} - 4$, *respectively.*

Proof. From Theorem 3.5, there exists a $(2^2, 2^4; 1, 3)$ PM over $GF(2^2)$ whose columns have zero-weight. The transpose of this PM is a $(2^4, 2^2; 3, 1)$ PM whose rows have zero-weight. We apply Theorem 4.1 once to the rows of this PM to get a $(2^{10}, 2^2; 3, 2)$ PM over $GF(2^2)$ whose columns have linear complexity at most $2^{10} - 2^4$. Applying Theorem 4.1 a further one, two, respectively, three times to the columns of this PM, we obtain PMs with the required parameters and having appropriate bounds on linear complexities of rows. ▮

Proof of Theorem 6.1 *for* $2 \leqslant u \leqslant 6$. We recall that the k of interest lie in the range of (6.1), and the definitions of t and s.

We begin with the case $u = 2$. Here we have $v = 2$ and $0 \leqslant t \leqslant \alpha + 1$. These values of t correspond to k with $\alpha \leqslant k \leqslant 2\alpha + 1$ and, since $u = v$, Theorem 6.1 imposes no conditions on the linear complexity of rows. We simply use Theorem 3.5 for $\alpha \leqslant k \leqslant 2\alpha - 1$. For $k = 2\alpha$, we use Result 6.2 and for $k = 2\alpha + 1$, a suitable PM can be obtained by transposing a PM from Theorem 3.5.

For $u=3$ and $v=2$, we have $0 \leqslant t \leqslant \alpha+1$. We consider three cases. For $t=0, 1$, PMs with the required parameters and bounds on linear complexity of rows can be obtained from Lemma 6.3 and Lemma 6.4, respectively. For $t=\alpha, \alpha+1$, the PMs of Lemma 6.5 and Lemma 6.6 have the required parameters and bounds on linear complexities of rows. In the final case we consider t with $2 \leqslant t \leqslant \alpha-1$. Here, $s=2$ and $k-\alpha s=2\alpha+t-2\alpha=t$ so that $2^{k-\alpha s}=2^t \geqslant s+1$. By (6.2), (6.3), and Theorem 4.2, there exists a $(2^{k-\alpha s}, 2^{2\alpha s-k}; s, 1)$ PM A over $GF(2^\alpha)$ with zero-weight rows. Applying Theorem 4.1 once to the rows we obtain a $(2^k, 2^{2\alpha s-k}; s, 2)$ PM B over $GF(2^\alpha)$ whose columns have linear complexity at most $2^k-2^{k-\alpha s}$. Since $s \geqslant u/2$ and $s+1 \leqslant 2^{k-\alpha s}$, we have $u-s \leqslant 2^{k-\alpha s}$. Thus we can make $u-s$ applications of Theorem 4.1 to the columns of B to obtain a $(2^k, 2^{6\alpha-k}; 3, 2)$ PM C over $GF(2^\alpha)$ whose rows have linear complexity at most $2^{6\alpha-k}-2^{4\alpha-k}$. Here $2^{4\alpha-k}$ is analogous to the expression $2^{\alpha(u-1)v-k}$ of (6.4) and so C has the required bound on the linear complexity of rows.

For $u=3$ and $v=3$, we have $1 \leqslant t \leqslant \alpha+1$. In the case $t=1$, a suitable PM is provided by Theorem 3.5 (as usual, we have no conditions on the linear complexity of rows when $u=v$). In the case $t=\alpha+1$, a suitable PM was constructed in Lemma 6.7. Otherwise, we have $2 \leqslant t \leqslant \alpha$, $s+1=3$, and $k-\alpha s=t$ so that $2^{k-\alpha s} \geqslant 4 > s+1$. After noting that (6.2), (6.3), and (6.4) hold, we can argue as in the remainder of the proof above to obtain suitable PMs.

Next we deal with $u=4$, $v=2$. We have $0 \leqslant t \leqslant \alpha+2$. For $t=0, 1$, suitable PMs were obtained in Lemma 6.8. We assume then that $t \geqslant 2$. In the case where $\alpha=2$ and $t=\alpha+2$, a suitable PM was found in Lemma 6.10. For all other cases, we calculate that $s=3$ and $k-\alpha s=t$. Then $2^{k-\alpha s}=2^t \geqslant 4=s+1$. Inequalities (6.2), (6.3), and (6.4) hold and we can again argue as in the proof above to obtain suitable PMs.

For $u=4$ and $v=3, 4$, we must consider $1 \leqslant t \leqslant \alpha+2$. For $t=1$ and $v=3$ suitable PMs were constructed in Lemma 6.9. For $t=1$ and $v=4$, PMs from Theorem 3.5 can be used, since we again have no conditions on the linear complexities of rows in this case. Otherwise $t \geqslant 2$, $s=3$, and $k-\alpha s=t$. Once again, $2^{k-\alpha s}=2^t \geqslant 4=s+1$. Inequalities (6.2), (6.3), and (6.4) hold and the usual argument applies.

For $u=5$, $v=2$, we have $0 \leqslant t \leqslant \alpha+2$. For $0 \leqslant t \leqslant \alpha-1$, $s=3$ and $k-\alpha s=\alpha+t$, so that $2^{k-\alpha s}=2^{\alpha+t} \geqslant 4=s+1$. Inequalities (6.2), (6.3), and (6.4) hold and we can use the usual argument to obtain suitable PMs. For $t=\alpha, \alpha+1, \alpha+2$, we have $s=4$ and $2^{k-\alpha s}=2^t$. Except in the case where $t=\alpha$ and $\alpha=2$, we have $2^{k-\alpha s} \geqslant s+1$ and the usual argument applies. The remaining case is covered in Lemma 6.10.

In the cases $u=5$, $v=3, 4, 5$, we have $1 \leqslant t \leqslant \alpha+2$. For $1 \leqslant t \leqslant \alpha$, we have $s=3$ and $2^{k-\alpha s}=2^{\alpha+t} \geqslant 8 > s+1$, while for $t=\alpha+1, \alpha+2$, we have $s=4$ and $2^{k-\alpha s}=2^t \geqslant 2^{\alpha+1} \geqslant 8 > s+1$. Again the usual argument can be applied.

For $u=6$, $v=2$, we have $0\leqslant t\leqslant \alpha+2$, $s=4$, and $2^{k-\alpha s}=2^{\alpha+t}$. Except in the case where $\alpha=2$ and $t=0$, we have $2^{k-\alpha s}\geqslant 8>s+1$ and the usual argument applies. In the exceptional case, a suitable PM is constructed in Lemma 6.10.

Finally, we consider the cases where $u=6$ and $3\leqslant v\leqslant 6$. We have $1\leqslant t\leqslant \alpha+2$ and $s=4$. Moreover, $2^{k-\alpha s}=2^{\alpha+t}\geqslant 8>s+1$ and the usual argument applies. ∎

7. Proof of Theorem 1.4.

We write $c=p^{\alpha}$, where p is a prime and $\alpha\geqslant 1$. For every u, $v\geqslant 1$ and each k with

$$\log_p(u+1)\leqslant k\leqslant \alpha uv-\log_p(v+1),$$

we aim to construct a p^{α}-ary $(p^k, p^{\alpha uv-k}; u, v)$ PM. When $p=2$ and $\alpha=1$, we can involve the result of [21]. Consequently we assume for the remainder of the proof that if $p=2$, then $\alpha\geqslant 2$. By transposing if necessary, we can also assume that $u\geqslant v$. Moreover, when $v=1$, a suitable PM has columns that are just the sequences of a (u, p^k, p^{α}) PF, where $u<p^k\leqslant p^{\alpha u}$. Such PFs were constructed in [22] (see Section 3), so we can further assume that $u\geqslant v\geqslant 2$.

We write $k=w+\alpha ut$, where $\log_p(u+1)\leqslant w<\alpha u+\log_p(u+1)$ and $0\leqslant t\leqslant v-1$, and we consider two cases: $0\leqslant t\leqslant v-2$ and $t=v-1$.

First, suppose that $0\leqslant t\leqslant v-2$. Then $2\leqslant v-t\leqslant u$ and by Theorem 5.1 in the case $p\geqslant 3$ and Theorem 6.1 in the case $p=2$, $\alpha\geqslant 2$, there exists a p^{α}-ary $(p^w, p^{\alpha u(v-t)-w}; u, v-t)$ PM A over $GF(p^{\alpha})$ whose rows have linear complexities less than or equal to $p^{\alpha u(v-t)-w}-(u-(v-t))$. Since $u-(v-t)\geqslant t$, the rows of A have linear complexities at least t less than $p^{\alpha u(v-t)-w}$. We can therefore apply Theorem 4.1 t times to the rows of A to obtain a $(p^{w+\alpha ut}, p^{\alpha u(v-t)-w}; u, v)$ PM over $GF(p^{\alpha})$. This PM has the desired parameters.

Second, consider the case $t=v-1$. Here $k=w+\alpha u(v-1)$, where we have $\log_p(u+1)\leqslant w\leqslant \alpha u-\log_p(v+1)$ in view of Result 1.1. We divide the possible values for w into two further ranges:

(i) When $\log_p(u+1)\leqslant w\leqslant \alpha u-\alpha v$, we have

$$\log_p(u+1)\leqslant w\leqslant \alpha u-\alpha v<\alpha(u-1)\qquad (\text{since } v\geqslant 2).$$

By Theorem 4.2 there exists a $(p^w, p^{\alpha u-w}; u, 1)$ PM A over $GF(p^{\alpha})$ whose rows have linear complexity at most $p^{\alpha u-w}-p^{\alpha(u-1)-w}$. Now

$$\alpha(u-1)-w\geqslant \alpha(u-1)-\alpha u+\alpha v=\alpha(v-1),$$

so that $p^{\alpha(u-1)-w} \geqslant p^{\alpha(v-1)} \geqslant v-1$. Thus we can make $v-1$ applications of Theorem 4.1 to the rows of A to obtain a $(p^{w+\alpha u(v-1)}, p^{\alpha u-w}; u, v)$ PM over $GF(p^\alpha)$. This PM has the desired parameters.

(ii) When $\alpha u - \alpha v + 1 \leqslant w \leqslant \alpha u - \log_p(v+1)$, we have

$$\log_p(v+1) \leqslant \alpha uv - k = \alpha u - w \leqslant \alpha v - 1,$$

and so by Theorem 3.5, there exists a p^α-ary $(p^{\alpha uv-k}, p^k; v, u)$ PM. The transpose of this PM has the desired parameters. ∎

Appendix A

We aim to use the results of [21] to show that there exists a (binary) $(2^k, 2^{\alpha u-k}; u, \alpha)$ PM with zero-weight columns for each $2 \leqslant u \leqslant 4$, $\alpha \geqslant 2$ and k with $2^u < 2^k < (u+1)\,2^u$, with the exception of the case $u=2$, $\alpha=2$, $k=3$. In fact we will construct PMs whose transposes have these parameters and properties. We consider each u in turn.

When $u=2$, we must have $k=3$:

(i) $\alpha=3$; for a $(2^3, 2^3; 3, 2)$ PM with zero-weight rows, see [21, Section 5].

(ii) $\alpha=4$; for a $(2^5, 2^3; 4, 2)$ PM with zero-weight rows, see [21, Appendix A, 1.1 (iii)].

(iii) $\alpha=5$; for a $(2^7, 2^3; 5, 2)$ PM with zero-weight rows, see [21, Appendix A, 2.1 (v)].

(iv) $\alpha \geqslant 6$; by [21, Lemma 6.1], there exists a $(2^{\alpha-3}, 2^3; \alpha, 1)$ PM whose rows have linear complexity 5. Applying Theorem 4.1 to the rows of this PM, we obtain a $(2^{2\alpha-3}, 2^3; \alpha, 2)$ PM whose rows have linear complexity 6 and, hence, zero-weight.

When $u=3$, we have $k=4$:

(i) $\alpha=2$; a $(2^2, 2^4; 2, 3)$ PM with zero-weight rows is A_1 of [21, Section 5].

(ii) $\alpha=3$; for a $(2^5, 2^4; 3, 3)$ PM with zero-weight rows, see [21, Section 5].

(iii) $\alpha=4$; for a $(2^8, 2^4; 4, 3)$ PM with zero-weight rows, see [21, Appendix A, 1.2 (vi)].

(iv) $\alpha=5$; for a $(2^{11}, 2^4; 5, 3)$ PM with zero-weight rows, see [21, Appendix A, 2.2 (vi)].

(v) $\alpha = 6$; by [21, Theorem 6.4], there exists a $(2^8, 2^4; 6, 2)$ PM whose rows have linear complexity at most $2^4 - 2^2 = 12$. Applying Theorem 4.1 to the rows of this PM, we obtain a $(2^{14}, 2^4; 6, 3)$ PM whose rows have linear complexity 13, and, hence, zero-weight.

(vi) $\alpha \geqslant 7$; by [21, Lemma 6.1], there exists a $(2^{\alpha-4}, 2^4; \alpha, 1)$ PM whose rows have linear complexity 9. Applying Theorem 4.1 to the rows of this PM twice, we obtain a $(2^{3\alpha-4}, 2^4; \alpha, 3)$ PM whose rows have linear complexity 11 and, hence, zero-weight.

When $u = 4$, we have $k = 5$ or $k = 6$. We begin with PMs for $u = 4$ and $k = 5$:

(i) $\alpha = 2$; a $(2^3, 2^5; 2, 4)$ PM with zero-weight rows is the transpose of the PM of [21, Appendix A, 1.1 (iii)].

(ii) $\alpha = 3$; a $(2^7, 2^5; 3, 4)$ PM with zero-weight rows is the transpose of the PM of [21, Appendix A, 1.2 (iii)].

(iii) $\alpha = 4$; we construct a $(2^{11}, 2^5; 4, 4)$ PM following the scheme of [21, Appendix A, 1.3]. We begin with the $(2^7, 2^5; 4, 3)$ PM of [21, Appendix A, 1.2 (v)] whose rows have linear complexity at most $2^5 - 2^3 + 2$. We apply Theorem 4.1 to the rows to obtain a PM with the desired parameters whose rows have linear complexity at most $2^5 - 2^3 + 3$ and, hence, zero-weight.

(iv) $\alpha = 5$; for a $(2^{15}, 2^5; 5, 4)$ PM with zero-weight rows, see [21, Appendix A, 2.3 (vii)].

(v) $6 \leqslant \alpha \leqslant 8$; by [21, Theorem 6.4], there exists a $(2^{2\alpha-5}, 2^5; \alpha, 2)$ PM whose rows have linear complexity at most $2^5 - 2^3$. Applying Theorem 4.1 to the rows of this PM twice, we obtain a $(2^{4\alpha-5}, 2^5; \alpha, 4)$ PM whose rows have linear complexity at most $2^5 - 2^3 + 2$ and, hence, zero-weight.

(vi) $\alpha \geqslant 9$; by [21, Lemma 6.1], there exists a $(2^{\alpha-5}, 2^5; \alpha, 1)$ PM whose rows have linear complexity 17. Applying Theorem 4.1 to the rows of this PM three times, we obtain a $(2^{4\alpha-5}, 2^5; \alpha, 4)$ PM whose rows have linear complexity 20, and, hence, zero-weight.

Finally we consider $u = 4$ and $k = 6$:

(i) $\alpha = 2$; a $(2^2, 2^6; 2, 4)$ PM with zero-weight rows is A_3 of [21, Section 5].

(ii) $\alpha = 3$; a $(2^6, 2^6; 3, 4)$ PM with zero-weight rows is the transpose of the PM of [21, Appendix A, 1.2 (iv)].

(iii) $\alpha = 4$; we construct a $(2^{10}, 2^6; 4, 4)$ PM following the scheme of [21, Appendix A, 1.3]. We begin with the $(2^6, 2^6; 4, 3)$ PM of [21, Appendix A, 1.2 (iv)] whose rows have linear complexity at most $2^6 - 2^3$.

We apply Theorem 4.1 to the rows to obtain a PM with the desired parameters whose rows have linear complexity at most $2^5 - 2^3 + 1$ and, hence, zero-weight.

(iv) $\alpha = 5$; for a $(2^{14}, 2^6; 5, 4)$ PM with zero-weight rows, see [21, Appendix A, 2.3 (vii)].

(v) $\alpha = 6$; by [21, Theorem 6.3], there exists a $(2^6, 2^6; 6, 2)$ PM whose rows have linear complexity at most $2^6 - 2^4$. We apply Theorem 4.1 twice to the rows of this PM to obtain a PM with the desired parameters whose rows have linear complexity at most $2^6 - 2^4 + 2$ and, hence, zero-weight.

(vi) $7 \leqslant \alpha \leqslant 9$; by [21, Theorem 6.4], there exists a $(2^{2\alpha-6}, 2^6; \alpha, 2)$ PM whose rows have linear complexity at most $2^6 - 2^4$. Applying Theorem 4.1 to the rows of this PM twice, we obtain a $(2^{4\alpha-6}, 2^6; \alpha, 4)$ PM whose rows have linear complexity at most $2^5 - 2^4 + 2$ and, hence, zero-weight.

(vii) $\alpha \geqslant 10$; by [21, Lemma 6.1], there exists a $(2^{\alpha-6}, 2^6; \alpha, 1)$ PM whose rows have linear complexity 33. Applying Theorem 4.1 to the rows of this PM three times, we obtain a $(2^{4\alpha-6}, 2^6; \alpha, 4)$ PM whose rows have linear complexity 36 and, hence, zero-weight.

Appendix B

In the notation following Result 3.2, we take $(s) = [001220211]$, a span 2 de Bruijn sequence over $GF(3)$ and

$$(a) = (d) = [001021122031654782307357725064387152634681$$
$$-428337624185367051755661327458604084488],$$

a span 2 de Bruijn sequence over $\mathbf{Z}_9$ (obtained from Construction 5.1 of [20] generalised to arbitrary alphabet sizes). We define an array A over $GF(3)$ by taking column 0 of A to be (s) and for $i \geqslant 1$, column i of A to be (s) shifted by a_{i-1} relative to column $i-1$. A is then a 3-ary $(3^2, 3^4; 2, 3)$ PM over $GF(3)$ whose columns clearly have zero-weight. That the rows of A have zero-weight was verified by computer program.

Appendix C

We give a construction for a $(2^7, 2^5; 3, 2)$ PM over $GF(2^2)$ whose rows and columns all have zero-weight. We use and ad hoc generalisation of [9], Construction 5.5.

Let

$$(s)^0 = [0, 0, 0, 1], \qquad (s)^1 = [0, 1, 1, 1]$$

be the sequences of a $(3, 2^2, 2)$ PF over $GF(2)$. According to Lemma 4.1 of [22], the sixteen sequences over $GF(2^2)$ with components of the form

$$((s)^i, E^k(s)^j), \qquad 0 \leqslant i, j \leqslant 1, \quad 0 \leqslant k < 4,$$

form a $(3, 2^2, 2^2)$ PF Γ over $GF(2^2)$. Here, as usual, we write the elements of $GF(2^2)$ as binary 2-tuples. We derive a new PF Γ' from Γ by substituting the sequences $((s)^0, E^2(s)^1)$ and $((s)^0, (s)^1)$ by their shifts $E((s)^0, E^2(s)^1)$ and $E^3((s)^0, (s)^1)$. We denote the sequences of Γ' by $(t)^0, (t)^1, \ldots, (t)^{15}$. It is easy to check that Γ' has the property that

$$\sum_{i=0}^{15} t_j^i = (1, 1) \qquad \text{for each} \quad 0 \leqslant j < 4, \tag{C.1}$$

the sum being over $GF(2^2)$.

We form a $(2^4, 2^2; 1, 3)$ PM A over $GF(2^2)$ using the sequences of Γ' as the rows. From (C.1), every column of A has weight (1,1).

Now let (s) be the span 2 de Bruijn sequence over $GF(2^2)$ with component sequences, written one below the other for convenience,

$$\begin{bmatrix} 0000011011011011 \\ 0011001100110011 \end{bmatrix}$$

((s) was derived using Construction 5.1 of [20]) and consider the sequences of $(E-1)^{-1}(s)$ whose generating cycles begin with (0, 0) and (0, 1):

$$\begin{bmatrix} 0000001001001001 \\ 0001000100010001 \end{bmatrix}, \qquad \begin{bmatrix} 0000001001001001 \\ 1110111011101110 \end{bmatrix}.$$

From the properties of $(E-1)^{-1}$ the length-3 subsequences of this pair are all distinct and if (u_0, u_1, u_2) is a length-3 subsequence of one, then $(u_0+(1, 1), u_1+(1, 1), u_2+(1, 1))$ is a subsequence of neither. Moreover, these sequences share a common length-2 subsequence of terms, namely $((0, 0), (0, 1))$, and so they can be joined to form a sequence (d) of period 32,

$$(d) = \begin{bmatrix} 00000010010001001001001000001001 \\ 00010001000111011101110111010001 \end{bmatrix},$$

having the properties that

(i) the length-3 subsequences of (d) are all distinct,

(ii) if (u_0, u_1, u_2) is a length-3 subsequence of (d), then $(u_0+(1,1), u_1+(1,1), u_2+(1,1))$ is not.

Let $(d)'$ denote the sequence of period 31 obtained from (d) by deleting the leading term $(0,0)$ of (d) and define (a) to have generating cycle that is the concatenation of four terms $(0,0)$ and four copies of the generating cycle for $(d)'$. Let $(e)^j$ denote column j of the $(2^4, 2^2; 1, 3)$ PM A. We define a new array B by defining column j of B to be the sequence of $(E-1)^{-1}(e)^j$ whose generating cycle begins with a_j.

We claim that B is a $(2^5, 2^7; 2, 3)$ PM with zero-weight rows and columns. The transpose of B has the properties we require. To verify our claims, note first that the columns of A have period 16, weight $(1,1)$, and, hence, linear complexity 16. Theorem 2.5 applies to show that each column of B has period 32, linear complexity 17, and, hence, zero-weight. Moreover, row zero of B is just (a) while the other rows of B are the cumulative sum of (a) with the rows of A. It is then easy to see that the rows of B all have period 128 and zero-weight. To show that B is PM one uses the properties of sequence (a) and those of $(E-1)^{-1}$ outlined in Theorem 2.5, Lemma 2.6, and Lemma 2.7. This requires a little more work, and we leave the details to the reader.

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